$\mathbb{E}(D_k \mid \mathcal{F}_{k-1}) = 0$

$\mathbb{E}(Y_k - Y_{k-1} \mid \mathcal{F}_{k-1}) \overset{?}{=} 0$

Martingale w.r.t $\mathcal{F}_k$

$\underbrace{\mathbb{E}(Y_k \mid \mathcal{F}_{k-1})}_{Y_{k-1}} - \underbrace{\mathbb{E}(Y_{k-1} \mid \mathcal{F}_{k-1})}_{Y_{k-1}} = 0$

(martingale)

(expec. & Y_k measurable w.r.t. $\mathcal{F}_k$ $\forall k$)

$\mu_{D_k \mid \mathcal{F}_{k-1}} = 0$

$$\left[\begin{array}{l} D_k = Y_k - Y_{k-1} \\ D_{k+1} = Y_{k+1} - Y_k \end{array}\right.$$

$\mathbb{E}(Y) = \mathbb{E}_X(\mathbb{E}(Y \mid X))$

$$D_k \overset{?}{\in} [a_k, b_k]$$

$$P(X \neq 1.5) = \cancel{0}\ 1$$

$$D_k = Y_k - Y_{k-1}$$ ← Doob's Construction

$$= \mathbb{E}\left(f(X_1, \dots X_n) \mid \underbrace{X_1 \dots X_k}_{X_1 \dots X_{k-1}\ \underbrace{x}_{X_k}}\right) - \underbrace{\mathbb{E}\left(f(X_1, \dots X_n) \mid X_1, \dots X_{k-1}\right)}_{c}$$

0 1

$$A_k := \inf_x D_k \qquad B_k := \sup_x D_k$$

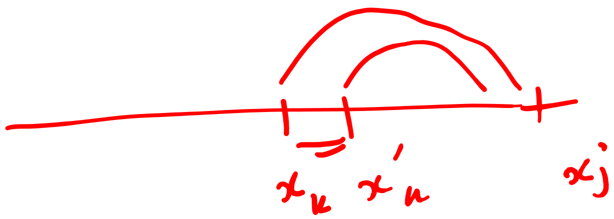
$$D_k - A_k = \mathbb{E}\left(f(X_1, \dots X_n) \mid X_1, \dots X_{k-1} X_k\right) - \inf_x \mathbb{E}\left(f(X_1, \dots X_n) \mid X_1, \dots X_{k-1}^{\,x}\right)$$

$$P(D_k - A_k > 0) = 1$$

$$f(x_1 \dots x_n) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu) \qquad X_i \in (a, b)$$

$$|f(x_1 \dots x_k \dots x_n) - f(x_1 \dots x'_k \dots x_n)| = \left|\tfrac{1}{n}(x_k - x'_k)\right|$$

$$\leq \tfrac{1}{n}(b-a)$$



$$f(x_1 \dots x_n) = \frac{1}{\binom{n}{2}} \sum_{i<j} g(x_i, x_j) \qquad g(x_i, x_j) = |x_i - x_j|$$

$$|f(x_1 \dots x_k \dots x_n) - f(x_1 \dots x'_k \dots x_n)| = \frac{1}{\binom{n}{2}} \Big(\sum_{j>k} g(x_k, x_j) - \sum_{j>k} g(x'_k, x_j) + \sum_{i<k} g(x_i, x_k) - \sum g(x_i, x'_k) \Big|$$

$$\leq \frac{(n-1)\,\overbrace{|x_k - x'_k|}^{(b-a)}}{\binom{n}{2}} = \frac{2(b-a)}{n}$$

$$\mathbb{E}\big(f(X_1 \cdots X_n) \mid \underbrace{X_1}_{x_1} \cdots \underbrace{X_{k-1}}_{x_{k-1}}, \underbrace{X_k}_{x_k}\big) \overset{\text{claim}}{=} \mathbb{E}_{X_{k+1}^n}\big(f(x_1 \cdots x_{k-1}\, x_k, \underbrace{X_{k+1} \cdots X_n}))$$

$$X_{k+1}^n \perp\!\!\!\perp X_1 \cdots X_k$$

$$B_k - A_k = \sup_x \mathbb{E}_{X_{k+1}^n}\big(f(x_1 \cdots x_{k-1}\, x, X_{k+1} \cdots X_n)\big) - \inf_x \mathbb{E}_{X_{k+1}^n}\big(f(x_1 \cdots x_{k-1}\, x\, X_{k+1}^n)\big)$$

$$\leq \sup_{x,y} \mathbb{E}\big(\underbrace{|f(x_1 \cdots x_{k-1}\, x, X_{k+1}^n) - f(x_1 \cdots x_{k-1}\, y\, X_{k+1}^n)|}_{\leq L_k}\big) \leq L_k$$

if f satisfies bounded-diff. prop. $\Rightarrow$ $B_k - A_k \leq L_k$

$$P\left(\left|\sum_{k=1}^{n} D_k\right| > t\right) \leqslant 2e^{-2t^2 / \sum_{k=1}^{n} L_k^2}$$

$$\boxed{\text{f has bounded diff. prop.}}$$

$$P\left(\left|\underbrace{Y_n}_{f(X_1,\dots X_n)} - \underbrace{Y_0}_{\mathbb{E}(f(X_1,\dots X_n))}\right| > t\right)$$

$$P\left(\left|U - \mathbb{E}(U)\right| > t\right) \leqslant 2e^{-\frac{\overset{n}{\cancel{2}}t^2}{\underset{2}{\cancel{4}}(b-a)^2}}$$

McDiarmid's Ineq.